

nanoSoC Test Board Datasheet

SoC Labs

July 11, 2025

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1 Introduction

2 Summary

3 FPGA Connections

4 Communications interface

4.1 I2C

Device	Identifier	Adress	Purpose
CDCM6208	U1	10101(AD1 AD0)	Clock Controller
INA228	U2	1000000	VDDACC Monitor
INA228	U3	1000001	VDD CORE Monitor
INA228	U4	1000100	VDDIO Monitor

5 References

Bugblat PIF uses raspberry pi to program the Mach XO2 Chip: PIF link

FPGA Pin Name	FPGA Pin Number	FPGA Bank	Nanosoc Pin Name	Nanosoc Pin Number
PL16A	25	3		
PL16B	26	3		
PL17A	27	3		
PL17B	28	3		
PL18D	31	3	SE	
PL19A	32	3	TEST	
PL19B	33	3	P0[0]	
PL20A	34	3	P0[1]	
PL20B	35	3	P0[2]	
PB3A	38	2	P0[3]	
PB3B	39	2	P0[4]	
PB4A	40	2	P0[5]	
PB4B	41	2	P0[6]	
PB7A	42	2	P0[7]	
PB7B	43	2	P0[8]	
PB9A	44	2	P0[9]	
PB9B	45	2	P0[10]	
PB10A	47	2	P0[11]	
PB10B	48	2	P0[12]	
PB13A	49	2	P0[13]	
PB13B	50	2	P0[14]	
PB15A	52	2	P0[15]	
PB15B	54	2	P1[0]	
PB20A	55	2	P1[1]	
PB20B	56	2	P1[2]	
PB21A	57	2	P1[3]	
PB21B	58	2	P1[4]	
PB23A	59	2	P1[5]	
PB23B	60	2	P1[6]	
PB24A	61	2	P1[7]	
PB24B	62	2	P1[8]	
PB24D	63	2	P1[9]	
PB27A	65	2	P1[10]	
PB27B	67	2	P1[11]	
PB29A	68	2	P1[12]	
PB29B	69	2	P1[13]	
PB30A	70	2	P1[14]	
PB30B	71	2	P1[15]	

Table 1: FPGA Connections to nanoSoC

FPGA Pin Name	FPGA Pin Number	FPGA Bank	Header Pin Name	Header Pin Number
PT28D	109	0	SA3	
PT28C	110	0	SA2	
PT27B	111	0	SA1	
PT27A	112	0	SA0	
PT25B	113	0		
PT25A	114	0		
PT24B	115	0		
PT24A	117	0		
PT23D	119	0		
PT23C	120	0		
PT21B	121	0		
PT21A	122	0		
PT20D	125	0	SDA	
PT20C	126	0	SCL	
PT18B	127	0		
PT18A	128	0		
PT15D	130	0		
PT15C	131	0		
PT14B	132	0		
PT14A	133	0		
PT13D	136	0		
PT13C	137	0		
PT11B	138	0		
PT11A	139	0		
PT10B	140	0		
PT10A	141	0		
PT9B	142	0		
PT9A	143	0		
		2		
		2		
		2		
		2		
		2		
		2		
		2		
		2		
		2		
		2		

Table 2: FPGA to Pi Header Connections

FPGA Pin Name	FPGA Pin Number	FPGA Bank	PMOD Pin
PL3A	1	5	8
PL3B	2	5	4
PL4A	3	5	7
PL4B	4	5	3
PL7A	9	5	6
PL7B	10	5	2
PL8A	11	5	5
PL8B	12	5	1

Table 3: FPGA to PMODA Connections